Agentic AI HVAC Bid Intelligence & Compliance Copilot

Project Brief | IBM SkillsBuild AI Experiential Learning Lab · Team MultiBasket

In one line: use AI to interpret complex bid requirements, deterministic software to verify what can be verified, and humans to retain control over high-risk decisions.

01 Background: from document generation to compliance intelligence

Our team first explored a browser-based AI assistant for recovering from failed digital tasks such as login, checkout, and online form errors. We quickly realized that the problem was too broad and that we did not have enough domain expertise to validate the real workflow, stakeholders, and operating constraints. Rather than spending weeks building around assumptions, we chose to pivot early.

The new direction came from first-hand experience with HVAC construction bidding. One team member had previously worked on bid-document preparation, and we followed that experience with interviews and questionnaires involving people who currently prepare bids. The feedback was consistent: the highest-value problem is not simply writing or formatting a bid document. The harder work is understanding what the solicitation actually requires, checking technical specifications, locating supporting evidence, and catching missing or non-compliant items before submission.

02 Product focus and core capabilities

We are designing a human-in-the-loop copilot for HVAC contractor bid teams. Instead of replacing bid staff, the system turns long, inconsistent solicitation documents into a structured compliance checklist, then helps gather evidence, run verifiable checks, and surface the items that need human judgment.

- Extract and classify qualification, personnel, scoring, documentation, and technical requirements from unstructured solicitation documents.
- Match qualification and staffing requirements against company records and personnel eligibility data.
- Compare technical requirements with manufacturer manuals, data sheets, and other supporting documents.
- Classify each requirement as Satisfied, Requires Human Review, Missing, or Non-Compliant.
- Link conclusions back to source evidence so bid staff can verify the result instead of relying on a black-box AI answer.

03 Why AI helps - and why AI alone is not enough

Bid documents vary widely in wording and structure. The same business requirement can appear in very different language across projects, which makes fixed rules and keyword matching insufficient for reliable interpretation. AI is useful for understanding and routing these requirements. But whenever correctness can be enforced explicitly, the final check should be handled by deterministic software rather than model judgment.

AI / Agent	Deterministic Services	Human Review
Interpret natural-language clauses; extract and classify requirements; decide which tool or data source to use; identify uncertainty and escalate.	Query structured records; validate certificate dates and staff eligibility rules; perform numerical parameter checks; verify file existence and assemble structured outputs.	Handle incomplete, conflicting, or high-risk cases; retain control over pricing, signatures, final approval, and formal submission.

04 A concrete example: let AI find it, let code verify it

Suppose a solicitation requires COP >= 5.8, while the manufacturer data sheet lists COP = 5.62 for the proposed unit. AI can identify the relevant requirement in one document and the corresponding specification in another. But the actual comparison should be done by deterministic code. The system should then flag the mismatch and send it for human or manufacturer follow-up, rather than allowing the model to reinterpret a clear numerical conflict.

05 Typical workflow

1	**Interpret the requirement**	The agent reads a clause and determines whether it concerns qualifications, personnel, scoring, technical specifications, or documentation.
2	**Choose a verification path**	Based on the requirement type, the system selects the appropriate company record, personnel database, manufacturer document, or comparison tool.
3	**Retrieve evidence**	The system gathers the relevant certificate, staff record, technical specification, or supporting document.
4	**Run the check**	Rules and calculations that can be verified explicitly are executed by deterministic services rather than left to model judgment.
5	**Assign a status**	The result is labeled Satisfied, Requires Human Review, Missing, or Non-Compliant, with the supporting evidence attached.
6	**Escalate risk**	Conflicts, missing information, or high-risk decisions stop the automated flow and are routed to bid staff for review.

06 MVP scope and value

The first MVP focuses on five things: requirement extraction, evidence matching, technical-parameter compliance checks, missing-item detection, and human review. Pricing decisions, electronic signatures, final project-manager assignments, final approval, and formal submission remain deliberately outside the automated workflow.

- Reduce repetitive reading, searching, and cross-checking across long solicitation documents.
- Surface mandatory and high-risk requirements earlier, before they become last-minute submission issues.
- Make technical-specification checks more consistent and traceable.
- Connect each requirement to the company or manufacturer evidence that supports it.
- Flag missing, unsupported, or non-compliant items while keeping business-critical decisions under human control.

07 Current stage

The team has completed initial problem validation, user research, and MVP scoping. We have established our GitHub repository, Agile workflow in ZenHub, and a Python/FastAPI backend. An initial tender PDF upload endpoint has been implemented, while document validation, requirement extraction, evidence matching, and agent orchestration remain under development. Our next milestone is to establish a reliable tender-document processing pipeline before integrating AI-based requirement analysis and compliance checking.